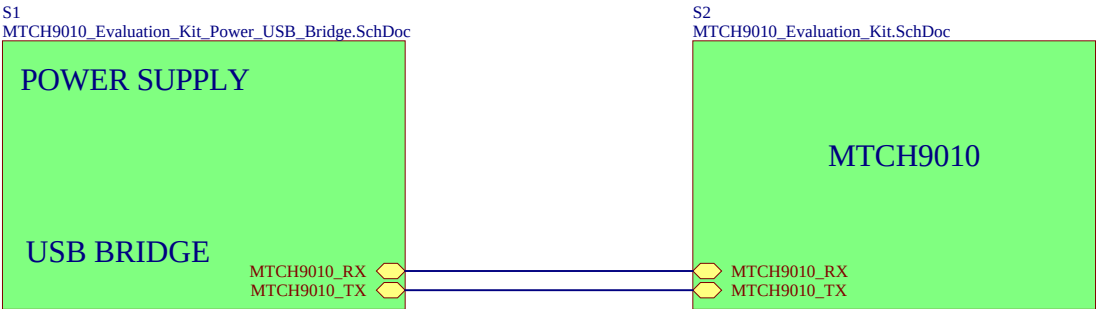


MTCH9010 Evaluation Kit



SCR1
Phillips Screw 1/4"

SCR2
Phillips Screw 1/4"

SCR3
Phillips Screw 1/4"

SCR4
Phillips Screw 1/4"

STANDOFF1
1/4" Standoff Nylon

STANDOFF2
1/4" Standoff Nylon

STANDOFF3
1/4" Standoff Nylon

STANDOFF4
1/4" Standoff Nylon

Test1
\$ > _

TEST, MTCH9010 Evaluation Kit

LABEL1

[ASSY# / REV]
SN: [SERIAL]
[DATE yyyy.mm.dd]

PCBA LABEL 18X6mm

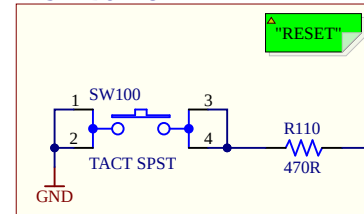
"Text in Silkscreen on PCB"				CNANO Template Revision: 2.8				
Project Owner: SLT								
PCB Layout Contact: SLT								
PartNumber: EV24U22A		Project Title MTCH9010 Evaluation Kit		Variant: Default Assembly				
Sheet Title Top Level						Designed with  Altium.com		
Size A3	SCH #: 02-01228		Rev: 5		Date: 2025-04-14			
	PCB #: 04-12292		Rev: 5		Sheet 1 of 4			
File: MTCH9010_Evaluation_Kit_TopLevel.SchDoc								

S3
MTCH9010_Evaluation_Kit_Revision_History.SchDoc



MTCH9010 Evaluation Kit

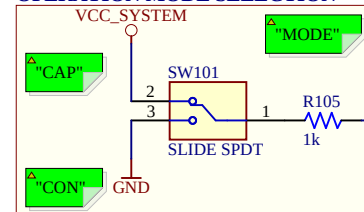
RESET BUTTON



RESET

▲ Use the reset button to latch new board configuration.

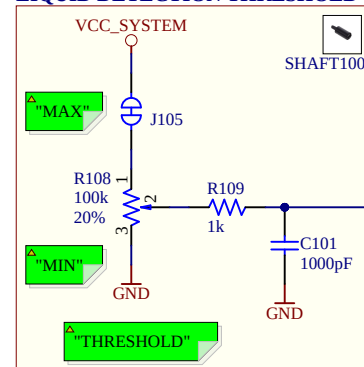
OPERATION MODE SELECTION



OPERATION_MODE

△ Select between capacitive or conductive operation. Pin setting can be overridden using Enhanced Configuration.

LIQUID DETECTION THRESHOLD

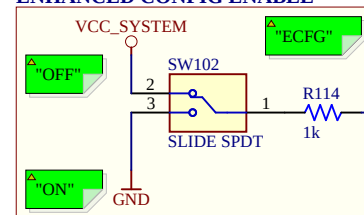


THRESHOLD

▲ Detection threshold: lowest at MIN, highest at MAX. Threshold value can be overridden using Enhanced Configuration.

△ Detection threshold maximum can be adjusted by soldering a resistor to the cut-strap J105.

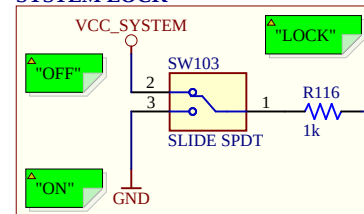
ENHANCED CONFIG ENABLE



ECFG ENABLE

Enable or disable Enhanced Configuration Mode.

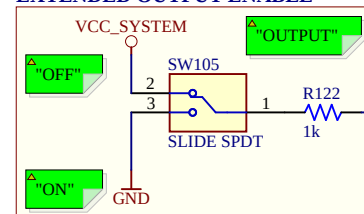
SYSTEM LOCK



SYSTEM LOCK

Save configuration to NVM; requires Enhanced Configuration Mode.

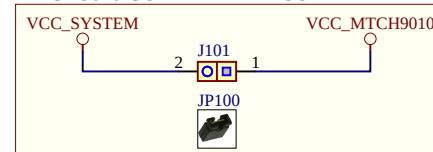
EXTENDED OUTPUT ENABLE



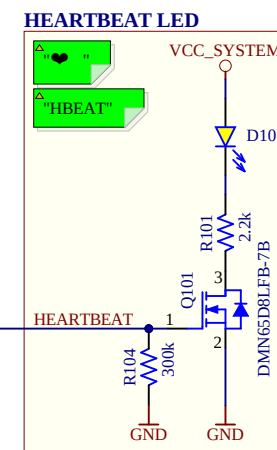
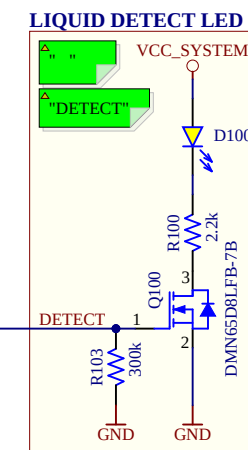
OUTPUT ENABLE

△ Enable or disable Extended Output. Pin setting can be overridden using Enhanced Configuration.

MTCH9010 CURRENT MEASUREMENT



▲ To measure the current consumption of the MTCH9010, remove JP100 and connect an ammeter to J101.

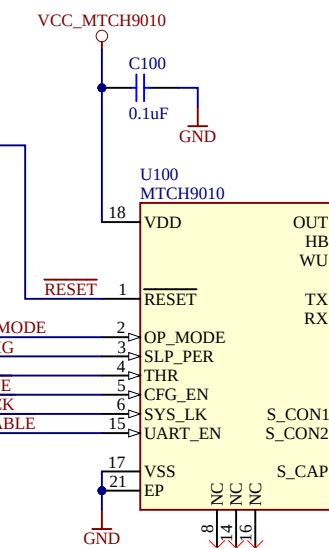


EXTERNAL CONTROL INTERFACE



△ The J100-7 pin can connect external circuitry to VCC_SYSTEM or receive external power. It is labeled "VDD" in silkscreen.

CAUTION: Power pins on J100 do not have polarity protection. Remove jumper from J201 when supplying external power.



"MTCH9010"

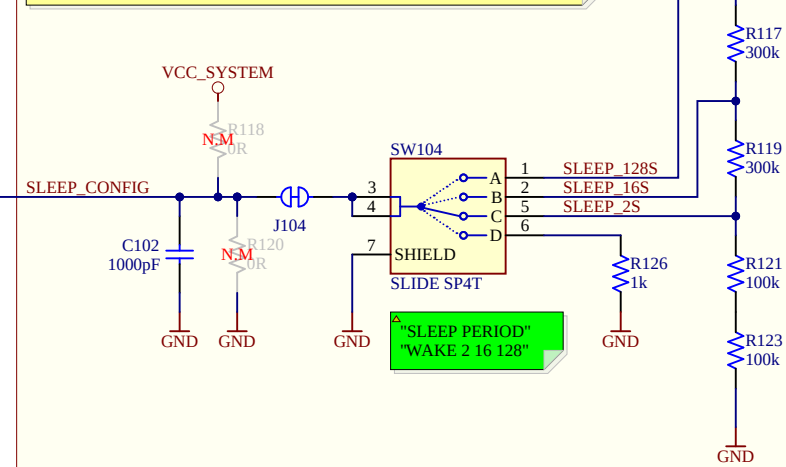
SLEEP PERIOD CONFIGURATION

△ The slide switch allows selecting a preset sleep period or waking the MTCH9010 by an external WAKE-UP signal.

Preset sleep intervals: 2s, 16s, 128s

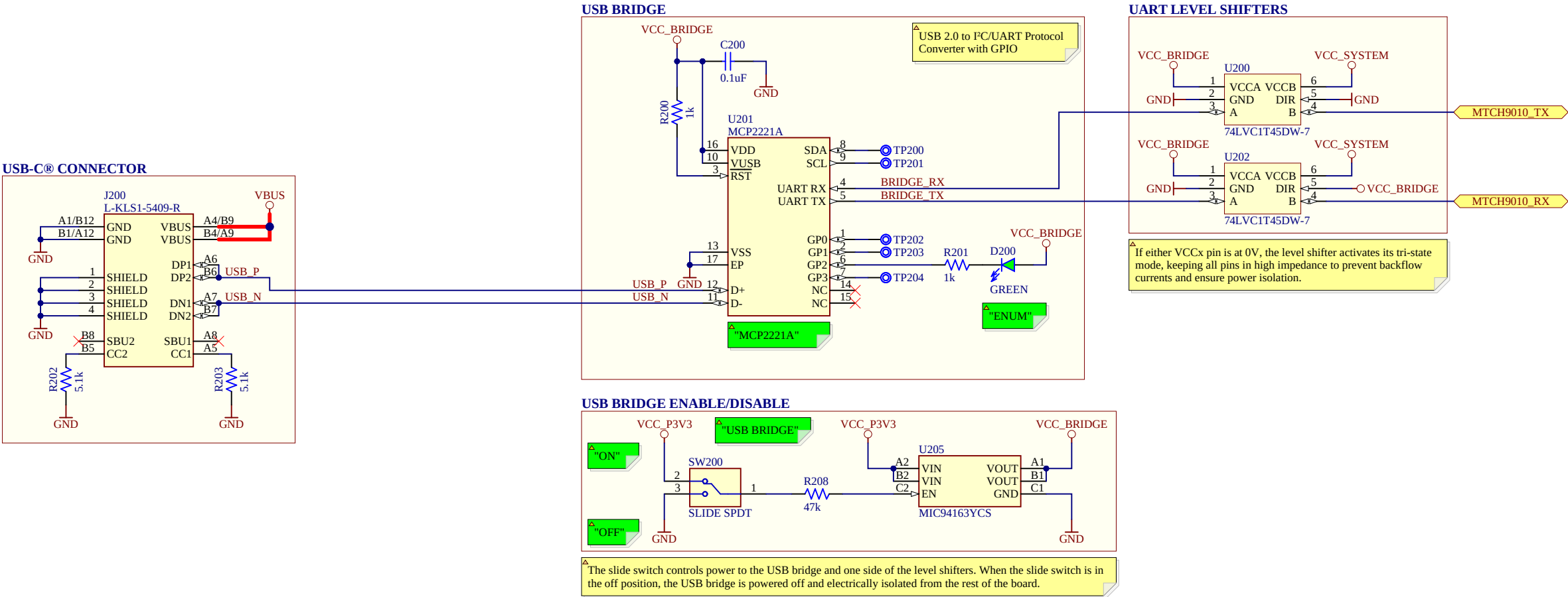
User-configurable sleep: Cut J104 a

Sleep settings can be overridden using Enhanced Configuration.

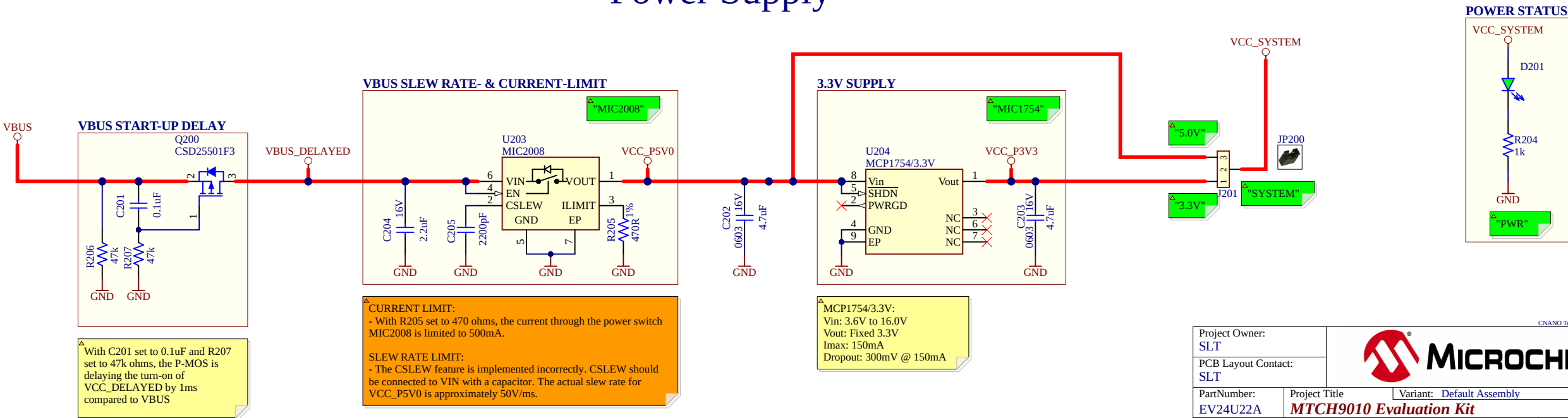


Project Owner: SLT		 MICROCHIP		C/NANO Template Revision: 2.8	
PCB Layout Contact: SLT					
PartNumber: EV24U22A		Project Title MTCH9010 Evaluation Kit		Variant: <u>Default Assembly</u>	
Sheet Title MTCH9010 Evaluation Kit		Designed with  Altium.com			
Size A3					
SCH #: 02-01228		Rev: 5		Date: 2025-04-14	
PCB #: 04-12292		Rev: 5		Sheet 2 of 4	
File: MTCH9010 Evaluation Kit.SchDoc					

USB Bridge



Power Supply



Revision History

PCB Assembly Rev 4:

Design Changes:

Initial Design

PCB:

PCB revision 4

PCB Assembly Rev 5:

Design Changes:

- Added capacitor and series resistor to reset circuit

- Changed Enhanced configuration switch connections

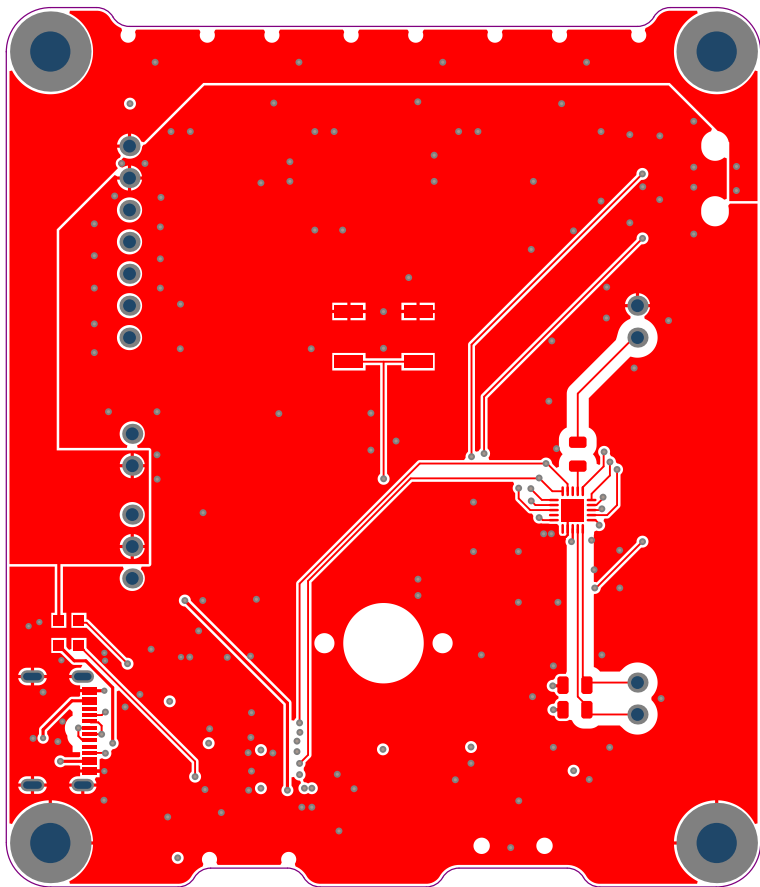
- Changed detection threshold potentiometer connections

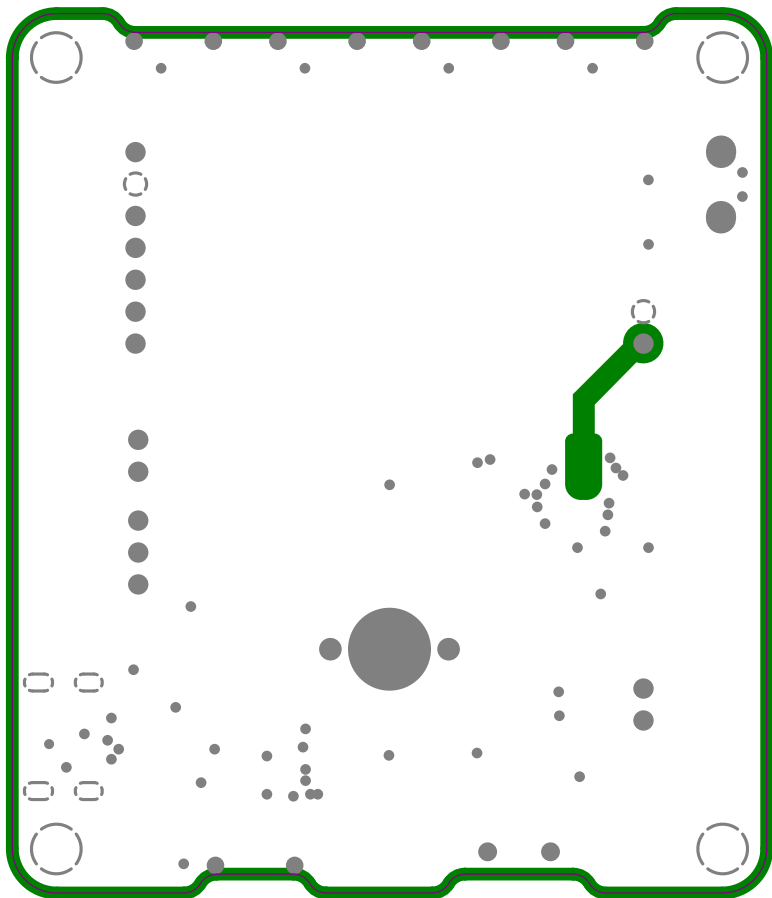
- Added cut-strap to allow users to limit detection threshold maximum

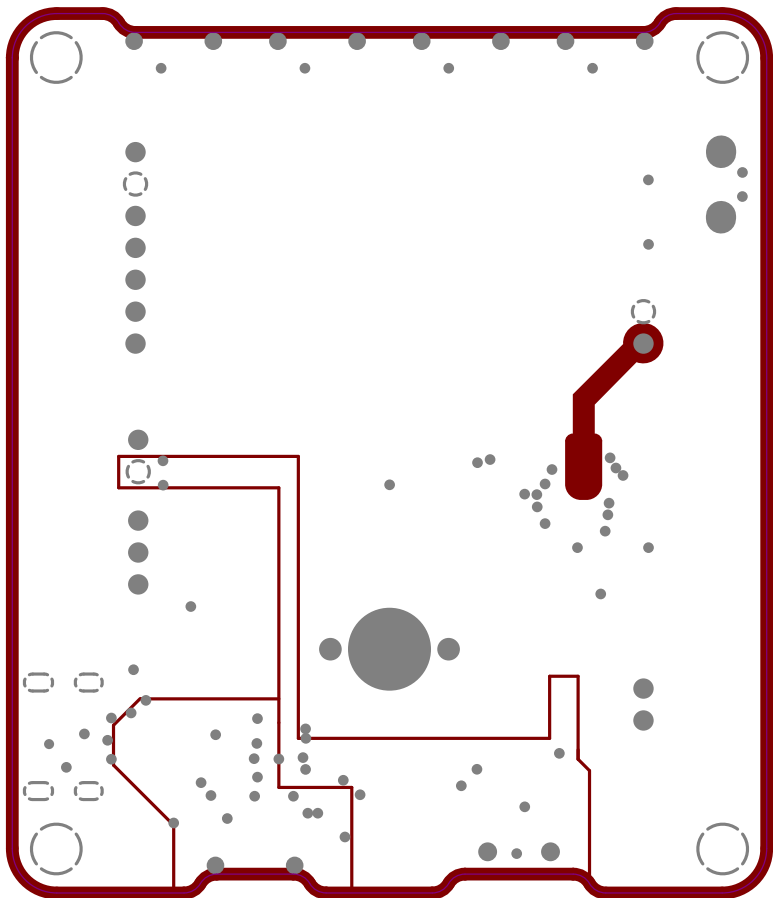
PCB:

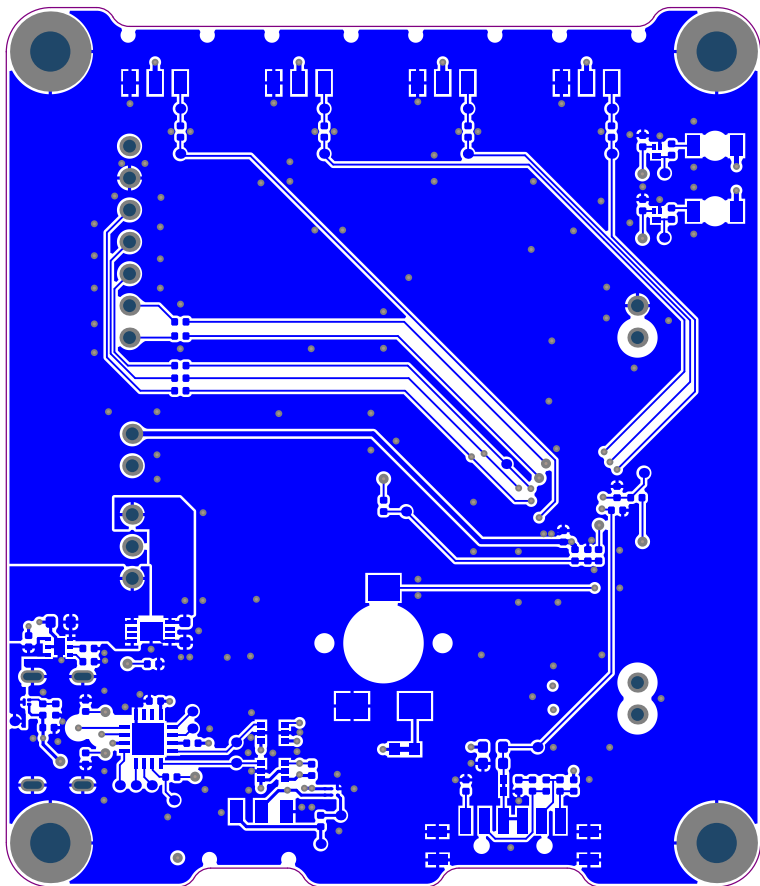
PCB revision 5

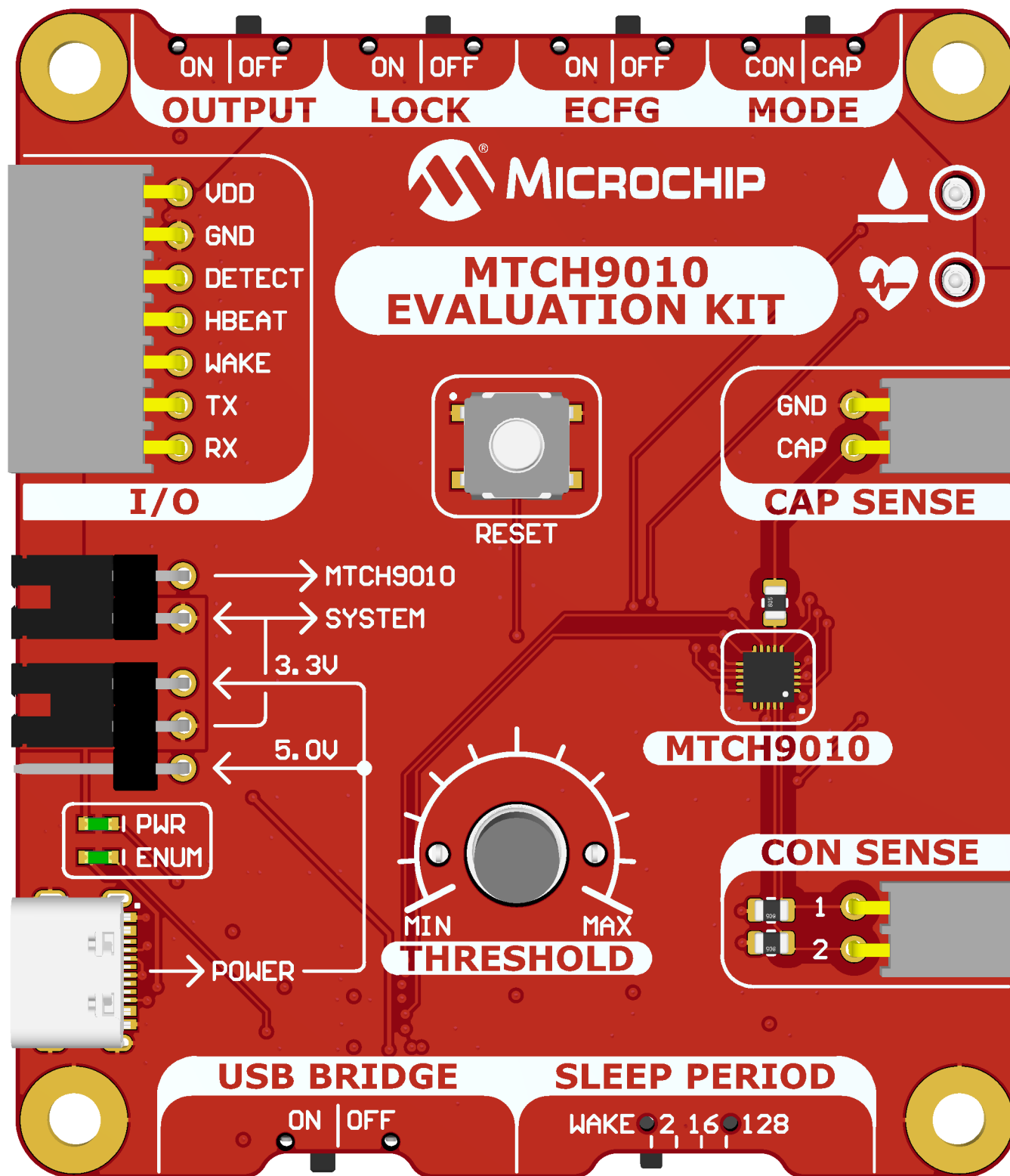
CNANO Template Revision: 2.8











ON | OFF
OUTPUT

ON | OFF
LOCK

ON | OFF
ECFG

CON | CAP
MODE



**MTCH9010
EVALUATION KIT**

I/O

- VDD
- GND
- DETECT
- HBEAT
- WAKE
- TX
- RX

RESET

CAP SENSE

- GND
- CAP

MTCH9010

CON SENSE

- 1
- 2



THRESHOLD

USB BRIDGE

ON | OFF

SLEEP PERIOD

WAKE 2 16 128

DETECT

HBEAT

04-12292 Rev5



[ASSY# / REV]
SN: [SERIAL]
[DATE yyyy.mm.dd]

Microchip © 2025

VDD

GND

DETECT

HBEAT

WAKE

TX

RX

GND

CAP

UK
CA



MTCH9010

SYSTEM

3V3

SYSTEM

5V0

MCP1754

MIC2008

MCP2221A

GP0
GP1

GP3
SDA
SCL

UCC_BRIDGE

CON1

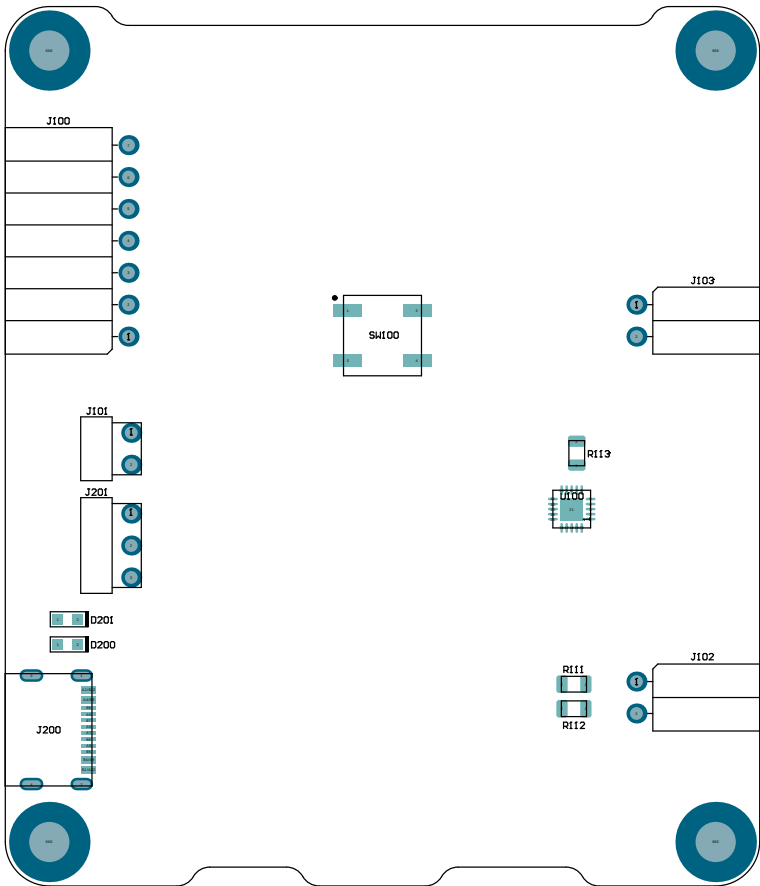
CON2

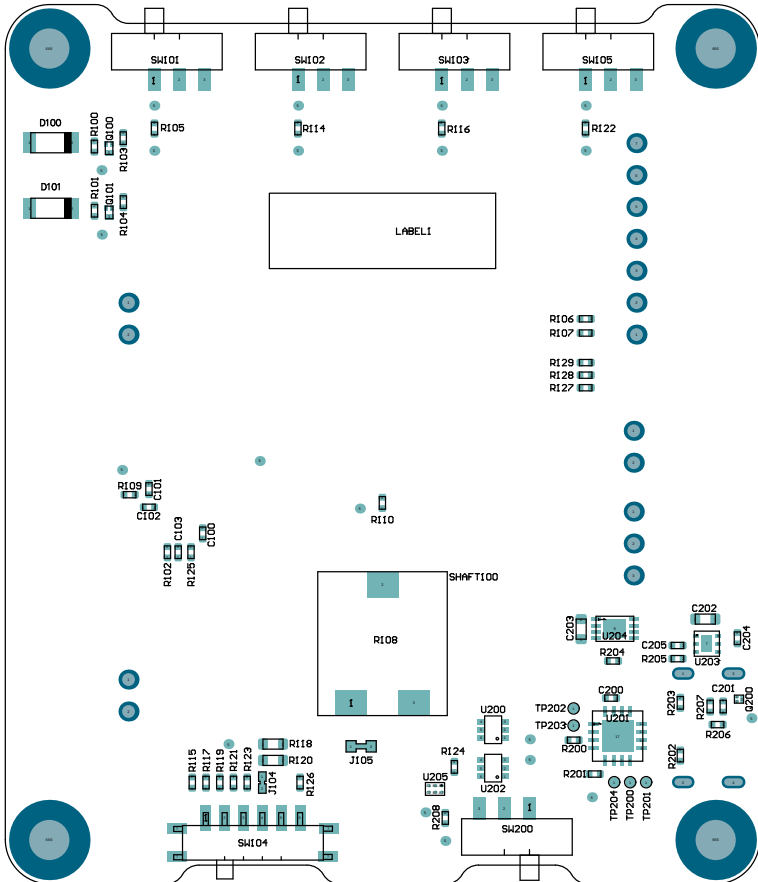
R118
R120

MIC94163

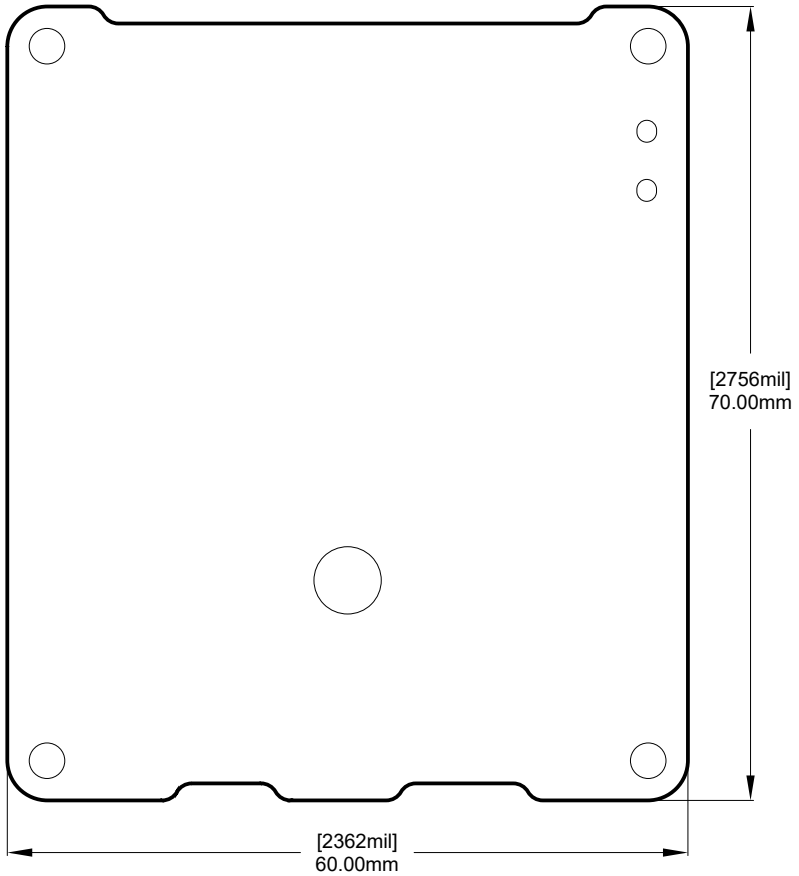
J105

J104

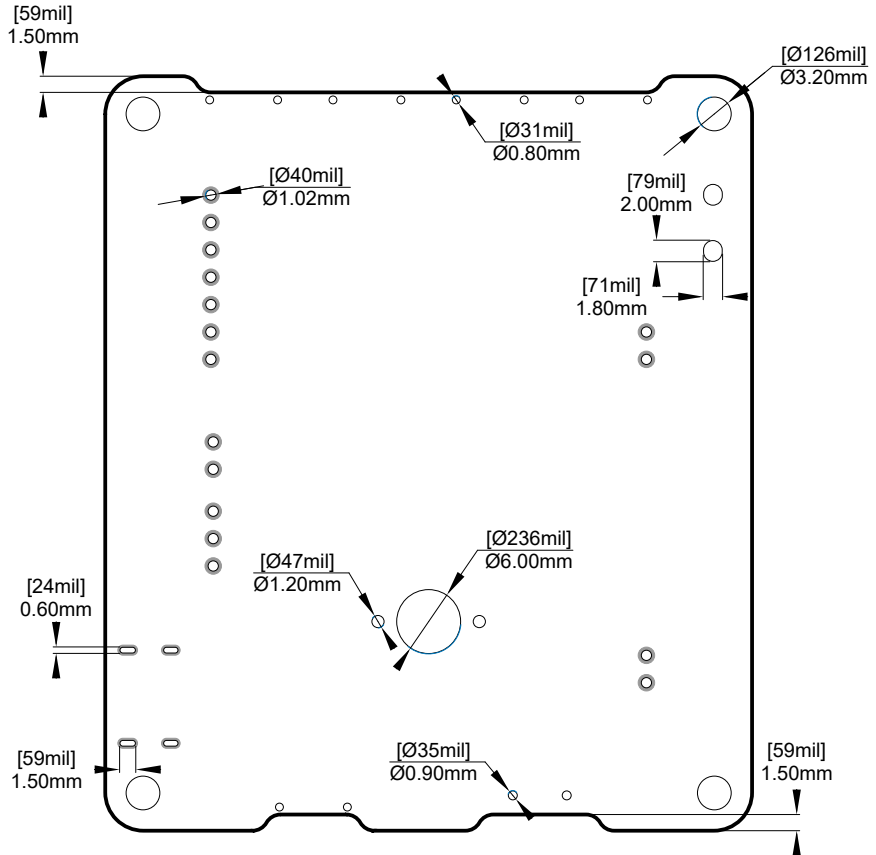




Board Outline



Through-Hole Dimensions



PCB Stackup

Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.010mm	Solder Resist	Solder Mask	GTS
Copper	Top Layer	0.035mm		Signal	GTL
Prepreg		0.117mm	PP-016	Dielectric	
Copper	Gnd Plane 1	0.035mm		Internal Plane	GP1
Core		1.200mm	FR-4	Dielectric	
Copper	Gnd Plane 2	0.035mm		Internal Plane	GP2
Prepreg		0.117mm	PP-016	Dielectric	
Copper	Bottom Layer	0.035mm		Signal	GBL
Surface Material	Bottom Solder	0.010mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO
Total thickness: 1.594mm					

Component list

BOM for Variant [Default Assembly] of Project [MTCH9010_Evaluation_Kit.PrjPcb] (No PCB Document Selected)

Source Data From: MTCH9010_Evaluation_Kit.PrjPcb
Project: MTCH9010_Evaluation_Kit.PrjPcb
Variant: Default Assembly



Report Date: 2025-04-14 14:13
Print Date: 2025-04-14 14:12:28

Fitted	Designator	Quantity	Value	#Column Name Error: Manufacturer	#Column Name Error: MPN	Description
Fitted	C100, C200, C201	3	0.1uF			
Fitted	C101, C102	2	1000pF			
Fitted	C103	1	10nF			
Fitted	C202, C203	2	4.7uF			
Fitted	C204	1	2.2uF			
Fitted	C205	1	2200pF			
Fitted	D100, D101	2	YELLOW			LED
Fitted	D200, D201	2	GREEN			LED
Fitted	J100	1	HDR-2.54 Female 1x7			
Fitted	J101	1	HDR-2.54 Male 1x2			
Fitted	J102, J103	2	HDR-2.54 Female 1x2			
Fitted	J200	1	USB2.0 Type-C FEMALE			
Fitted	J201	1	HDR-2.54 Male 1x3			
Fitted	JP100, JP200	2	JUMPER 2.54mm 1x2			
Fitted	LABEL1	1	PCBA LABEL 18X6mm			
Fitted	Q100, Q101	2	DMN65D8LFB-7B			Transistor MOS
Fitted	Q200	1	CSD25501F3			Transistor MOS
Fitted	R100, R101	2	2.2k			Resistor
Fitted	R102, R110, R205	3	470R			Resistor
Fitted	R103, R104, R117, R119	4	300k			Resistor
Fitted	R105, R106, R107, R109, R114, R116, R122, R126, R127, R128, R129, R200, R201, R204	14	1k			Resistor
Fitted	R108	1	100k			Resistor Var
Fitted	R111, R112, R113	3	10k			Resistor
Fitted	R115, R121, R123	3	100k			Resistor
Fitted	R124, R206, R207, R208	4	47k			Resistor
Fitted	R125	1	10k			Resistor
Fitted	R202, R203	2	5.1k			Resistor
Fitted	SCR1, SCR2, SCR3, SCR4	4	Phillips Screw 1/4"			
Fitted	SHAFT100	1	MISC PT10-SHAFT			
Fitted	STANDOFF1, STANDOFF2, STANDOFF3, STANDOFF4	4	1/4" Standoff Nylon			
Fitted	SW100	1	TACT SPST			Switch
Fitted	SW101, SW102, SW103, SW105, SW200	5	SLIDE SPDT			
Fitted	SW104	1	SLIDE SP4T			
Fitted	Test1	1				
Fitted	U100	1	MTCH9010			
Fitted	U200, U202	2	74LVC1T45DW-7			
Fitted	U201	1	MCP2221A			
Fitted	U203	1	MIC2008			
Fitted	U204	1	MCP1754/3.3V			
Fitted	U205	1	MIC94163YCS			
Not Fitted	R118, R120	0	0R			Resistor

Approved	87	Notes